Warehouse onboarding

Replace the 2-hour induction with 12 short exercises on the floor. Project name: Kata.

Sector: Enterprise learning, warehouse logistics

Status: concept study, written and designed by Karthik Bindumon, 2026. Numbers are targets or modelled assumptions and are labelled as such.

Method: The Realisation Loop, v1 (see 00-realisation-loop.pdf).

1 Brief

New warehouse pickers receive a two-hour slide induction on day one. Errors, near-misses and injuries cluster in the first week, on the floor, where the slides never were. This study replaces the induction with a first week structured as twelve graded kata, performed on live shelf sections.

2 Audience and needs analysis

Primary audience: new pickers, mixed literacy, often on a temporary contract, learning a warehouse management system (WMS) scanner and a physical route at the same time. What they know: how to lift a box. What they lack: the site's route logic, the scanner's failure modes, and what to do when a pick is wrong. Constraints: supervisors have no spare time; peak season; high churn in week one.

Needs analysis method: WMS error logs for new starters by tenure day, near-miss reports for the last two cohorts, four shadowing shifts with a buddy, and interviews with two pickers who left in week one.

3 The realisation

> **Nobody learns a route from a slide. The aisle is the syllabus. Move the learning to the shelf and cut the induction to twenty minutes.**

Contrarian core: the compliance slides are the thing to cut. Keep the legally required safety content as a twenty-minute session, then put every other minute on the floor.

4 Learning objectives (targets)

By day 7, new pickers complete a 40-line order unassisted with pick accuracy at or above the site's 90-day-tenure average. By day 30, near-miss reports per 1,000 picks for new starters match the site median.

5 Design

Kata card deck. Twelve cards, one micro-skill each: scan, locate, reach, lift, confirm, pack, exception handling, short-pick, damaged goods, hazard zone, end-of-shift, teach-back. Each card fits in a pocket and has three stamp boxes: Shu, Ha, Ri.

Progression. Shu: perform with the buddy watching. Ha: perform alone, buddy checks the result. Ri: teach it to the next hire. A card is complete only at Ri. Shelf sections for practice are marked with coloured tape; colour sets difficulty (line count, exceptions seeded).

Implicit learning. The route is taught by the layout: practice sections are placed so the natural walk between kata is the site's real pick route. By day 3 the picker knows the route without having been told it.

Artefacts. Card deck, tape map, buddy guide (one page), stamp, WMS query for the evaluation metrics.

6 Theory map

Source	How it is used
Ericsson, deliberate practice	Short, targeted, immediate feedback from a buddy at the shelf.
Bjork, desirable difficulties	Twelve spaced retrievals across a week instead of one massed session.
Roediger and Karpicke, testing effect	Every kata is a retrieval attempt. There is no re-reading.
Sweller, cognitive load theory	One micro-skill per card. Scanner and route are never introduced in the same kata.
Csikszentmihalyi, flow	Tape colours raise challenge as skill rises across the week.
Lave and Wenger, legitimate peripheral participation	Ri (teach-back) moves the picker from trainee to member of the crew by day 7.
Shu-Ha-Ri	The progression itself, from Japanese martial pedagogy.

7 Evaluation plan (Kirkpatrick)

Level	Measure	When
1 Reaction	One tick box per card: useful, or not.	Each kata
2 Learning	Kata pass rates, first attempt against retry, from the stamped cards.	Daily, week 1
3 Behaviour	Pick accuracy and near-miss rate per 1,000 picks, from the WMS, new starters only.	Day 7 and day 30
4 Results	Time to full productivity; week-one incident rate; week-one attrition. This cohort against the last two.	Day 30 and 90

8 Pilot plan and risks

One site, one hiring cohort (10 to 20 people), one buddy per two starters. Risks: buddies cutting corners under peak load (mitigation: buddy hours are rostered, not volunteered); cards lost (spare deck at the desk; the WMS is the record of truth); starters treating Ri as optional (the teach-back is the graduation, publicly).

9 Ethics

Kata data belongs to the picker. Supervisors see pass or fail, never the number of attempts. The deck is not used in performance reviews.

10 What I would need from the client

WMS access for the four metrics, two buddies on roster, a tape budget, and agreement that the twenty-minute safety session stays the only classroom time.